A Novel Method to Remove Interference Fringes for Hyperspectral SWIR Imagers

Yue Xu, Shao-Biao Xie, Jian Liang, De-Xin Sun, Nai-Ming Qi, and Yin-Nian Liu, *Member, IEEE*

***Abstract*— A hyperspectral short-wave infrared (SWIR) imager with a HgCdTe focal plane detector usually suffers from interference fringes in spectral and spatial dimension due to the etalon effect. It can cause a modulation of sensitivity higher than 15% when the spectral resolution is higher than 10 nm. In this article, a two-step method is proposed and evaluated to remove interference fringes for hyperspectral SWIR imagers. First, we extract the spectral data, conduct harmonic decomposition to suppress the fringes in spectral dimension, and calculate the correction coefficients. Second, we optimize the correction coefficients according to the distribution of the interference fringes in spatial dimension. Sample SWIR images acquired from the Advanced Hyperspectral Imager (AHSI) on China's GaoFen-5 satellite are used to evaluate the performance of the algorithm. It shows that the proposed method can well preserve the original spectral pattern and reduce the peak-to-peak amplitude of spatial fringing from ±15% to ±4%. The method is compared with three other previously proposed methods. It turns out that the newly proposed method has strong adaptability, high accuracy, and high efficiency.**

***Index Terms*— Hyperspectral imaging, interference fringes, short-wave infrared (SWIR).**

I. Introduction

THE Advanced Hyperspectral Imager (AHSI), one of the six payloads on China's GaoFen-5 satellite launched on May 9, 2018, is a visible/short-wave infrared (SWIR) hyperspectral imager built by the Shanghai Institute of Technical Physics (SITP), Chinese Academy of Sciences (CAS) [1]. The SWIR imager of AHSI is a push-broom hyperspectral sensor with a total of 180 spectral channels ranging from 1000 to 2500 nm. The spectral resolution is higher than 10 nm, and the spatial resolution is higher than 30 m. Its signal is collected by a cooled HgCdTe focal plane detector, including four identical infrared focal plane arrays (IRFPAs) (512 × 512 pixel) with a staggering spatial arrangement as shown in Fig. 1. Each IRFPA has a four-layer internal structure, that is, ZnS, Si, CdTe, and HgCdTe layer from the top to bottom (Fig. 2), and the thickness of each layer is shown in Table I.

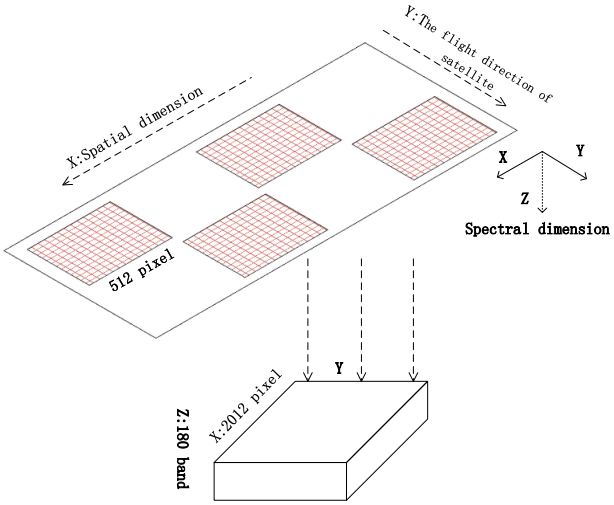

Fig. 1. SWIR HgCdTe detector of AHSI. (There are 36 overlapping pixels between the combined IRFPAs. The final spatial dimension of the SWIR detector is 2012 pixels.)

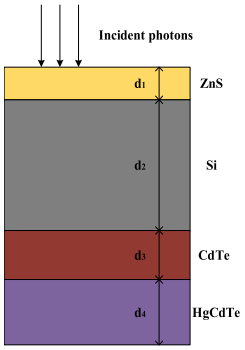

Fig. 2. Main internal multilayer structure of the IRFPAs.

TABLE I
THICKNESS OF EACH LAYER IN IRFPAS OF THE HgCdTe DETECTOR IN AHSI

Parameter	IRFPA1	IRFPA2	IRFPA3	IRFPA4
d_1	330nm	340nm	350nm	300nm
Estimated d_2	34.4μm	35.4μm	35.4μm	35.6μm
d_3	6.2μm	5.16μm	6.2μm	5μm
d_4	2.9~5.4μm	3.1~5.5μm	2.9~5.4μm	3~5.4μm

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The quantum efficiency of large IRFPA with Si substrate in the wavelength range of 1000–1100 nm is too low to meet the requirements of SWIR hyperspectral imaging. Therefore, the transmittance of the SWIR imager is increased by thinning the Si layer. Given the high SWIR spectral resolution (<10 nm) of AHSI, each pixel only receives extremely narrowband light. If the absorption length, that is, the path-length required for 1/e (e is the natural constant) of the incoming photons to be absorbed in silicon, is greater than the thickness of the Si layer in the HgCdTe detector, the incident light will interfere on the upper (ZnS) and lower (CdTe) surfaces, causing serious interference fringes known as the etalon effect [2]–[4]. For incident light with a wavelength of more than 1000 nm, its absorption length can be as high as 100 μm or even 1000 μm, which is tens of times of Si layer thickness (35 μm) [4]. As a result, interference fringes will occur in the SWIR image of AHSI, which brings difficulties for the subsequent processing and quantitative analysis of the hyperspectral data [5].

The interference fringes caused by the etalon effect are different from noises such as Gaussian noise, impulse noise, and dead pixels and lines. Actually, it affects both spectral and spatial domains with irregular patterns [4]. On one hand, the interference phase angle changes with wavelengths, resulting in different fringe patterns along the spectral dimension. On the other hand, local variations in the Si layer thickness may also change the interference phase angle, resulting in different fringe patterns along the spatial dimension. Furthermore, the fringes are heterogeneous in four IRFPA modules due to 1) the thickness variation in the Si layer during the manufacturing and 2) the possible different incidence angles caused by the placement error.

There are two conventional strategies to remove interference fringes. First, the IRFPA design can be improved, such as increasing the thickness of the Si layer [6], roughening the front surface of the Si layer [7], or adding graded antireflection (AR) coatings to match the wavelength for a specific band [8], [9]. However, increasing the thickness of the Si layer will sacrifice the quantum efficiency of the incident light at the wavelength of 1000–1100 nm; the roughening process of the front surface does not show significant improvement in suppressing fringes [10], and it is difficult to design graded AR coatings for a wide spectral range, which may largely increase the cost. Second, interference fringes can be reduced from postprocessing [10]–[14]. Malumuth *et al.* [11], [12] developed a thin-film model for the STIS-G750L spectral imager (visible and near infrared (VNIR) and VNIR bands) in the Hubble telescope to predict and correct fringing effects. This method requires detailed knowledge of the detector's physical structure, the thickness of each layer, central wavelength of the incident light received by each pixel, and the refractive index of each layer for different wavelengths. However, the acquisition of these parameters with sufficient precision is impossible considering the fabrication error or spectral shifting problem. Walsh *et al.* [13] proposed a method to correct the interference fringes in the STIS-G750L spectral imager (VNIR) based on the flat-field data. However, due to different imaging conditions and the wavelength shift problem, the fringing pattern between on-orbit data and calibration data could be different and lead to a poor defringing result. Rojo and Harrington [14] used wavelet analysis to remove interference fringes of images. It is an effective method to remove fringes on flat-field images, but its application to complex images does not work well if the interference fringes have inconsistent phase, frequency, and amplitude. In addition, algorithms such as block-matching and 4D filtering (BM4D) [15], low-rank matrix recovery (LRMR) [16], and group low-rank representation (GLRR) [17] are widely used in hyperspectral denoising. The LRMR method and GLRR method describe the problem of finding and exploiting low-dimensional structures in high-dimensional data. The assumption of low-rank matrix approximation algorithms is that some available high-quality image can be used as low-rank components. However, the distribution of the etalon effect on the hyperspectral images is regular and similar to texture, which can be wrongly put into low-rank components. The essence of the BM4D method is to find similar cubes in hyperspectral data. However, the patterns of the etalon effect are different on different bands and pixels, which makes it difficult to find similar cubes. A two-step method to remove fringes from dispersive hyperspectral VNIR imagers was proposed by Hu *et al.* [10]. It first uses a ridge regression model to suppress the spectral fringes, and then computes spatial correction coefficients based on the combination of spatial–spectral feature of hyperspectral data. Although it is an effective method, the spatial–spectral feature for SWIR data is usually weak due to the inconsistent enhancement or attenuation of the interference between adjacent bands.

In this article, a new two-step method to remove interference fringes is proposed to tradeoff between high quantum efficiency and high spectral resolution of SWIR hyperspectral imaging. The rest of this article is organized as follows. Section II shows fringes in the hyperspectral image data. Section III illustrates the details of the proposed method. Section IV gives a full evaluation of the proposed method in terms of image quality and spectral fidelity and compares it with other methods. Finally, conclusions are summarized in Section V.

II. Data

The raw data acquired by the SWIR imager of AHSI are a mosaic of four scenes from four IRFPA. The hyperspectral image data of East Tianshan in Xinjiang, China, were acquired by the SWIR imager on June 1, 2018. The image of East Tianshan data (band 94 1300 nm; band 1 60 2330 nm) with severe interference noise is shown in Fig. 3. The hyperspectral image data of North Dunhuang acquired on September 11, 2018 (band 94 160), are shown in Fig. 4. Both band 94 and band 160 are largely affected by interference fringes, and the fringes are heterogeneous in four IRFPA mosaic images. (More data can be seen in Figs. S1 and S2 in Supporting Information.)

III. Methods

The SWIR hyperspectral data cube acquired by AHSI can be expressed as $\mathrm{DN}_f(x, y, z)$, where x represents the cross-track spatial dimension, y denotes the along-track spatial dimension,

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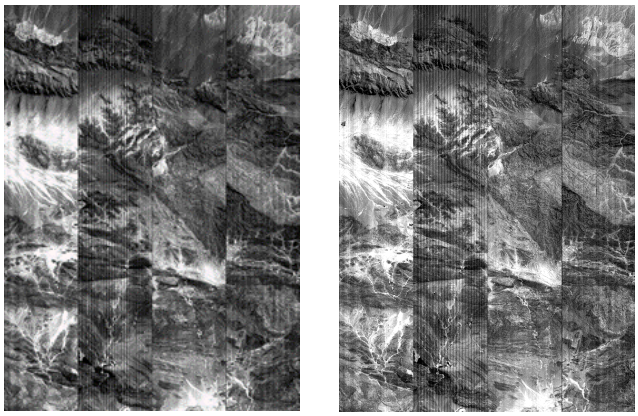

Fig. 3. East Tianshan data (a) (band 94) and (b) (band 160) acquired by AHSI.

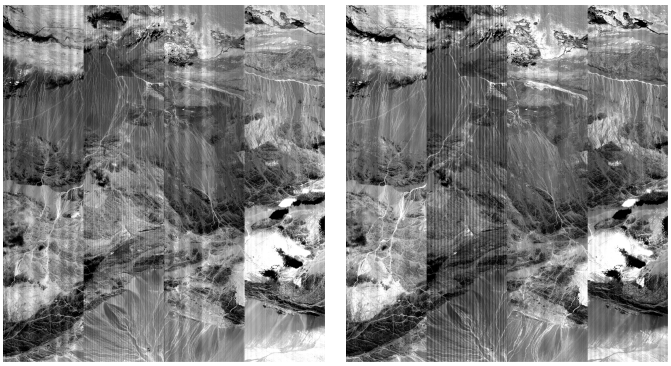

Fig. 4. (a) North Dunhuang data (a) (band 94) and (b) (band 160) acquired by AHSI.

and z represents the spectral dimension of 180 spectral channels from 1000 to 2500 nm. Because there are 12 overlapped pixels between the neighboring IRFPAs, the final x dimension has 2012 pixels (Fig. 1).

Hyperspectral data cube suffered from interference fringes can be expressed as

$$\mathrm{DN}_f(x, y, z) = \mathrm{Coef}_e * \mathrm{DN}_e(x, y, z) \tag{1}$$

where Coef_e is the fringing amplitude, DN_e is the expected response value or true value, and DN_f is the observed response value. The goal is to determine the coefficients for each pixel and estimate the corrected pixel response value as close as possible to the expected response value

$$\mathrm{DN}'_e(x, y, z) = \mathrm{Coef}_r * \mathrm{DN}_f(x, y, z) \approx \mathrm{DN}_e(x, y, z) \tag{2}$$

where Coef_r is the correction coefficient, and DN'_e is the corrected pixel response value.

A new two-step method based on harmonic decomposition is proposed to obtain correction coefficients. The first step is to use harmonic decomposition for *spectral* data, suppress the spectral fringes, and then compute correction coefficients. The second step is to fine-tune the correction coefficients according to the *spatial* distribution of the interference fringes in the image.

A. Step 1: Spectral Fringing Suppression

Given the high spectral resolution of 10 nm, the spectral responses of the adjacent channels are very similar, and the spectral signature curve is assumed to be smooth. However, the spectral curves suffer from interference fringes, that is, the incident light is modulated by interference enhancement or attenuation, resulting in high-frequency spikes in spectral curves. To obtain the true spectral curves, a harmonic decomposition model is used for spectral fringing suppression.

Harmonic decomposition was first proposed by Jakubauskas *et al.* [18], and it is especially effective in suppressing noise consisting of sine or cosine components. Because the fringes caused by interference are similar to sine modulation, the high-frequency spikes in the spectral curves can be denoised by harmonic decomposition [19]. In harmonic decomposition, the spectral curves are expressed as the superposition of sine and cosine waves in the forms of harmonic remainder, amplitude, and phase. Suppose that the spectral curve of each pixel is $F(k) = (f_1, f_2, \ldots, f_N)$, k is the serial number of bands, f represents the response value for each band, and N denotes the total number of bands, which is equal to 180 in this case. The harmonic remainder is set to A_0, the cosine amplitude is A_h, the sine amplitude is B_h, the harmonic component amplitude is C_h, and the phase of the harmonic component is θ_h. Then, the spectral curve can be expressed as

$$\begin{aligned} F(k) &= \frac{A_0}{2} + \sum_{h=1}^{\infty}\left[A_h \cos\left(\frac{2\pi hk}{N}\right) + B_h \sin\left(\frac{2\pi hk}{N}\right)\right] \\ &= \frac{A_0}{2} + \sum_{h=1}^{\infty}\left[C_h \sin\left(\frac{2\pi hk}{N} + \theta_h\right)\right] \end{aligned} \tag{3}$$

where h is the number of the harmonic decomposition ($h = 1, 2, 3, 4, \ldots$), and $(A_0/2)$, A_h, B_h, C_h, and θ_h are given by

$$\frac{A_0}{2} = \frac{1}{N}\sum_{k=1}^{N} f_k \tag{4}$$

$$A_h = \frac{2}{N}\sum_{k=1}^{N} f_k \cos\frac{2\pi hk}{N} \tag{5}$$

$$B_h = \frac{2}{N}\sum_{k=1}^{N} f_k \sin\frac{2\pi hk}{N} \tag{6}$$

$$C_h = \sqrt{\left(A_h^2 + B_h^2\right)} \tag{7}$$

$$\theta_h = \tan^{-1}\left(-\frac{A_h}{B_h}\right) \tag{8}$$

where harmonic remainder $(A_0/2)$ represents the mean response value of the ground object along the spectral dimension, C_h represents the response variation in the spectral value in each band, and θ_h represents the location of the band where the maximum response occurs.

The critical step for harmonic decomposition is to choose an appropriate number (h). Normally, a small harmonic number represents the main pattern of a spectral curve, and its amplitude and phase describe the general shape; a large number

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represents the local variation in a spectral curve, but possibly containing noises or abnormal changes. Therefore, if h is too small, the spectral curves will have an over-smoothing effect and lose spectral details, and if h is too large, the method cannot effectively remove the spectral fringes.

The spectral curves reconstructed by harmonic decomposition need further adjustments. For bands without interference fringes, there is little difference between the reconstructed spectral curves and the original ones. If bands after reconstruction have more than 25% difference, we believe that the spectral information have been recovered by removing interference fringes. Thus, the maximum change threshold is set to 25%.

After harmonic decomposition, the correction coefficients can be obtained for each pixel. The original image data are $\mathrm{DN}_f(x, y, z)$, and the data after harmonic decomposition are $\mathrm{DN}_g(x, y, z)$. Assume a linear response model, the gain correction coefficient is $\mathrm{Coef}_a(x, z)$, the offset correction coefficient is $\mathrm{Coef}_b(x, z)$, and the correction coefficient values along the y-direction are the same. The objective function is

$$\min\left(\mathrm{DN}_g(x, y, z) - \mathrm{Coef}_a(x, z) * \mathrm{DN}_f(x, y, z) + \mathrm{Coef}_b(x, z)\right)^2. \tag{9}$$

Through this linear regression model, the values of correction coefficients $\mathrm{Coef}_a(x, z)$ and $\mathrm{Coef}_b(x, z)$ can be obtained.

B. Step 2: Spatial Fringing Suppression

After harmonic decomposition in Step 1, we obtain the correction coefficients $\mathrm{Coef}_a(x, z)$ and $\mathrm{Coef}_b(x, z)$. In Step 2, we will fine-tune them according to the distribution of the interference fringes in the spatial dimension.

The spatial high-frequency oscillation effect caused by multibeam interference can be modeled by a function with a period of 2π. Assuming the fringe period in the spatial dimension is Δd, the incident light wavelength is λ, and the refractive index of the silicon layer is n_0. In a back-and-forth reflection process, we assume that the variation in the thickness of the Si layer causes a 2π phase difference in the spatial fringes' period [4]

$$4\pi n_0 \Delta d/\lambda = 2\pi. \tag{10}$$

Therefore, we have $\Delta d = \lambda/2n_0$.

The thickness variation in the Si layer in our IRFPA is within 3 μm. Using a sample AHSI image of East Tianshan, the spatial fringes' period of four IRFPAs in each band can be determined. Because the etalon effect occurs only when the wavelength range of the incident light is extremely narrow, the synthesized full band image from hyperspectral data can be considered free from interference fringes. Besides, the response characteristics of a pixel in one band can be represented by the mean value along the flight direction. Thus, the spatial fringes' amplitude distribution of a specific band can be obtained by the ratio of the mean value of the band to that of the full band along the flight direction

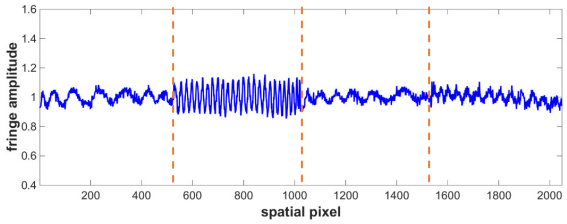

Fig. 5. Spatial fringes' distribution in SWIR band 160 of AHSI.

$$\mathrm{Fringe}(x, z) = \mathrm{mean}\left(\sum_{y=0}^{y_\max} \mathrm{DN}_f(x, y, z)\right) / \mathrm{mean} \times \left(\mathrm{mean}\left(\sum_{z=0}^{180}\sum_{y=0}^{y_\max} \mathrm{DN}_f(x, y, z)\right)\right) \tag{11}$$

where $y_\max$ is the maximum number of frames along the flight direction, $\mathrm{Fringe}(x, z)$ is the spatial amplitude distribution of the band z, and the values along the y-direction are the same.

A typical spatial fringes' distribution in SWIR band 160 of AHSI with severe interference noise is shown in Fig. 5, where the spatial fringe has a periodic pattern in each module. The number $t_i, i = 1, 2, 3, 4$ of spatial fringes' periods of the four IRFPAs is counted as $t_1 = 7$, $t_2 = 26$, $t_3 = 7$, and $t_4 = 13$, respectively. The number of spatial pixels is calculated by (12). $T_i (i = 1, 2, 3, 4)$ is the number of spatial pixels of each period and T is equivalent to T_i in the following equation:

$$T_i = \frac{512}{t_i}, \quad i = 1, 2, 3, 4. \tag{12}$$

The interference fringes in each IRFPA are regularly enhanced or attenuated but with several periodic variations in the spatial dimension, while the amplitudes of the adjacent periodic fringes are similar. Ignoring the fringes less than one cycle and the amplitude difference in the adjacent cycles, the correction coefficients in each IRFPA satisfy the statistical principle [10]

$$\mathrm{mean}\left(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_a(x, z)\right) \approx 1 \tag{13}$$

$$\mathrm{mean}\left(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_b(x, z)\right) \approx 0 \tag{14}$$

where M is the number of spatial fringes' periods determined by the algorithm and is set as 3, 15, 3, and 5 according to the period of interference fringes in four modules, respectively. The value of M takes into account the nonuniformity and the statistical characteristics of interference fringes. If the value of M is too small, the spatial dimension of the correction coefficient cannot meet the statistical characteristics of interference fringes due to nonuniformity. When the value of M of four modules is larger than or equal to 2, 10, 2, and 4, respectively, the correction shows satisfactory result and the difference between the correction coefficients is less than 0.002

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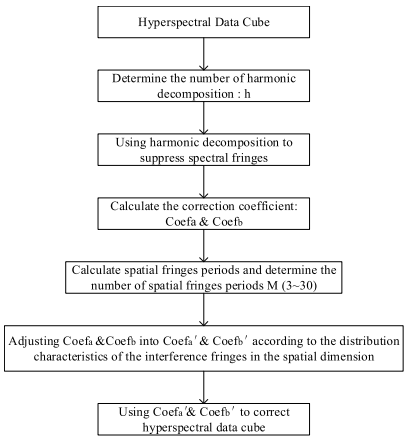

Fig. 6. Flowchart of the proposed method.

(see Figs. S3 and S4 in Supporting Information). After trial and error, the M values of 3, 15, 3, and 5 for four modules are selected.

In Step 1, there is overcorrection or insufficient correction, resulting in a value of mean$(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_a(x,z))$ significantly deviating from 1 and mean$(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_b(x,z))$ significantly deviating from 0. The corresponding correction coefficients are then updated by

$$\mathrm{Coef}_a{}'(x,z) = \mathrm{Coef}_a(x,z)/\mathrm{mean}\left(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_a(x,z)\right) \quad (15)$$

$$\mathrm{Coef}_b{}'(x,z) = \mathrm{Coef}_b(x,z) - \mathrm{mean}\left(\sum_{x=1}^{\mathrm{MT}} \mathrm{Coef}_b(x,z)\right). \quad (16)$$

By adjusting the statistical characteristics of the correction coefficients to match the statistical characteristics of the interference fringes in the spatial dimension, it effectively suppresses the over- or undercorrection caused by Step 1. The whole process of the proposed method is summarized in Fig. 6.

IV. Results and Discussion

The hyperspectral image data of East Tianshan are used to evaluate the effectiveness of the proposed method to remove the fringes. And then the obtained correction coefficients are transferred to a different scene in North Dunhuang, China, acquired on September 11, 2018, so as to evaluate the adaptability of the correction coefficients. Furthermore, the results are compared with that using the method proposed by Hu *et al.* [10].

A. Correction Results

1) Correction Results of the Data of East Tianshan: As mentioned in Section I, an appropriate choice of the number for harmonic decomposition should be determined according to the denoising effect and algorithm complexity. Fig. 7(a) and (b) are the spectral curves of ice and rock derived from East Tianshan with different harmonic decomposition numbers, that is, 15, 45, and 75 times, respectively.

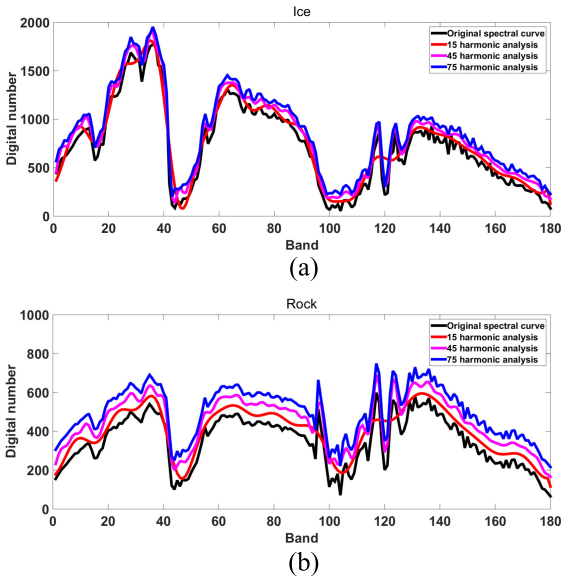

Fig. 7. Spectral curves of (a) ice and (b) rock before and after different numbers of harmonic decomposition.

The curves are shifted about 50, 100, and 150 digital numbers along the y-axis to avoid overlapping. As it can be seen, the uncorrected spectral curve (black line) suffers from obvious fringes; with a small decomposition number (15 times, red line), the spectral curve suffers from oversmoothing as a result of losing spectral details; with a large decomposition number (75 times, blue line), the spectral fringes cannot be well-removed. We used dozens of different ground features (lake, road, architecture, reed, etc.) to obtain a reasonable value of h (see Figs. S5 and S6 in Supporting Information). The root mean square (RMSE) and the mean curvature are used to determine the effectiveness of harmonic decomposition. RMSE represents the change in the spectral curves before and after harmonic decomposition, and the mean curvature represents the smoothness of the curve. The smaller the RMSE, the closer the two curves; and the smaller the mean curvature value, the smoother the corrected curve. The purpose of this step is to remove the fringes of the spectral curve while keeping the true spectral pattern, so we expect the RMSE and the mean curvature after correction to be small. The curves of RMSE and mean curvature with different harmonic decomposition numbers are shown in Fig. 8. When h is higher than 40, RMSE tends to be stable, which means the curve will not change significantly. When h is higher than 50, the mean curvature increase sharply which means the curve will not be smooth. Therefore, we believe that a value of 40–50 is reasonable. After trial and error, an appropriate decomposition number of 45 (pink line) is finally selected, and the original spectral curve can be well-preserved and the spectral fringes can be greatly suppressed.

The spectral curves of ice and rock before and after Step 2 correction are shown in Fig. 9. After correction, the spectral fringes have been largely suppressed, and the spectral shape before and after correction shows reasonable consistency. In addition, a spatial fringe comparison of before and after correction is shown in Fig.10, where the maximum

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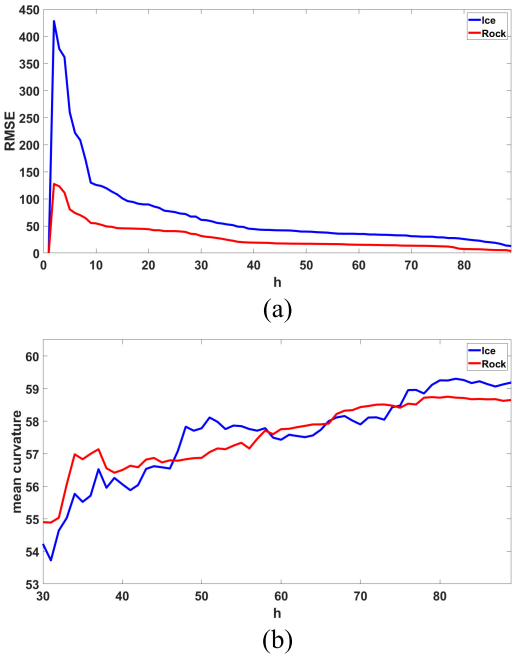

Fig. 8. (a) RMSE curves and (b) mean curvature curves with different numbers of harmonic decomposition. (Rock curve is shifted about 32 digital numbers along the y-axis.).

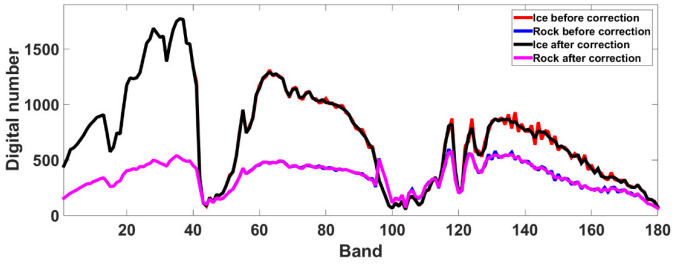

Fig. 9. Comparison of the spectral curves before and after correction.

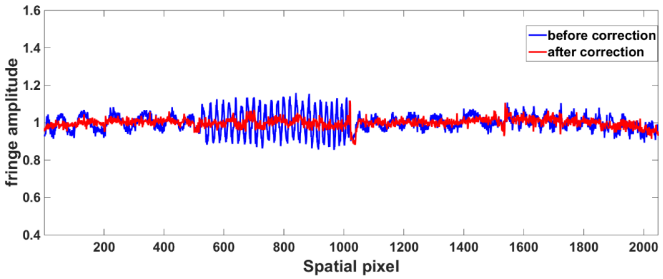

Fig. 10. Comparison of the spatial fringes before and after correction.

spatial fringes' amplitude is reduced from 15% to 4%. The images before and after harmonic decomposition (Step 1) and spatial correction (Step 2) in band 160 of the East Tianshan data are shown in Fig. 11(a)–(c), respectively. It can be clearly seen that the fringes have been largely suppressed after Step 1, and the more remaining fringes have been removed after Step 2.

After response equalization and stitching of the four IRFPAs, the false-color composite images (band 36 1300 nm, 94 1780 nm, and 160 2330 nm) of before and after correction

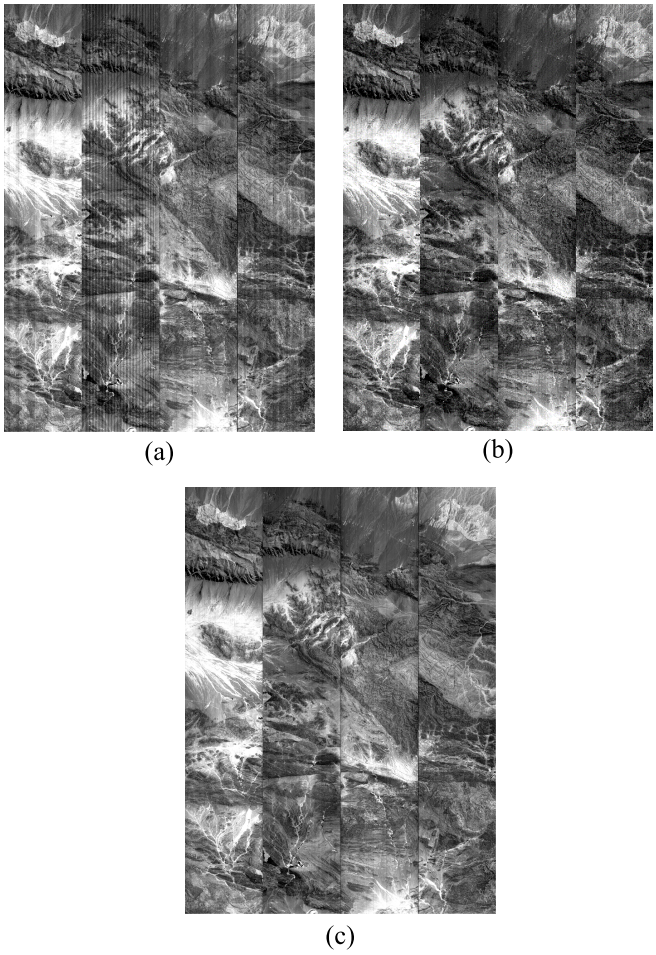

Fig. 11. East Tianshan data (band 160) acquired by AHSI. (a) Uncorrected hyperspectral image. (b) Hyperspectral image after Step 1 correction. (c) Hyperspectral image after Step 2 correction.

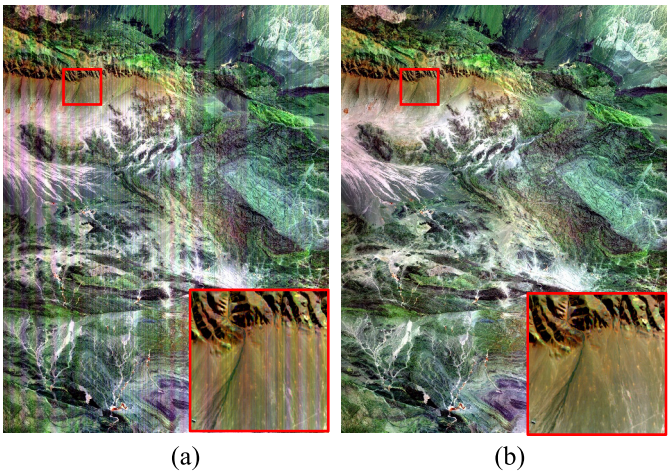

Fig. 12. False-color RGB image of East Tianshan. (a) Uncorrected hyperspectral image. (b) Corrected hyperspectral image using the method proposed by this article.

are shown in Fig. 12(a) and (b), respectively. From the enlarged insets, we can see that the original image suffers from severe fringes, and the corrected image is nearly free of them, providing a satisfactory result.

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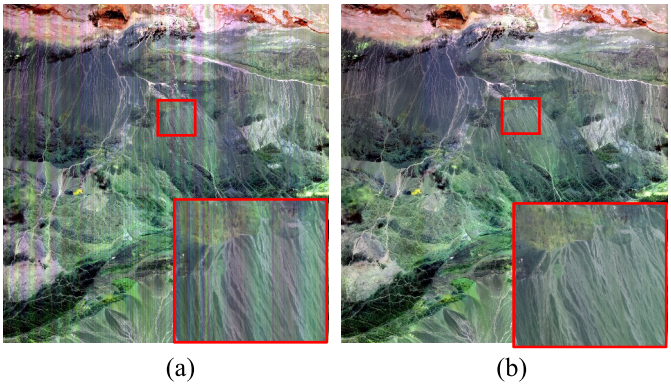

Fig. 13. False-color RGB image of the North Dunhuang data. (a) Uncorrected hyperspectral image. (b) Corrected hyperspectral image.

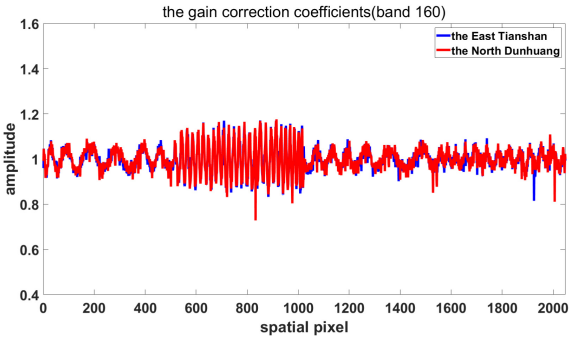

Fig. 14. Comparison of Coef_a obtained from different data.

2) Correction Results of the Data of North Dunhuang: To evaluate the adaptability and reliability of the proposed method, the correction coefficients obtained are transferred for a different scene in North Dunhuang acquired on September 11, 2018. The false-color composite images (band 36, 94, and 160) of before and after correction are shown in Fig. 13(a) and (b), respectively. It is shown that the correction coefficients obtained in East Tianshan can also effectively remove the interference fringes of the image in North Dunhuang, indicating the high adaptability of this method.

Fig. 14 is the comparison between the correction coefficient Coef_a obtained from the North Dunhuang data and the correction coefficient Coef_a obtained from the East Tianshan data. The two curves are almost the same, and the RMSE of the two is 0.202, indicating that the algorithm has high consistency and stability for the correction coefficients obtained from different scenes. Therefore, the correction coefficients obtained from North Dunhuang can also effectively remove the fringes of East Tianshan (the result can be seen in Fig. S7 in Supporting Information).

B. Comparison With Previous Method Based on Image Quality

The results of removing fringes by different methods are shown in Fig. 15. It can be clearly observed that the LRMR method can remove dead lines or small stripes but not interference fringes, and the BM4D method suppresses the interference fringes well in some module but losing a lot of details

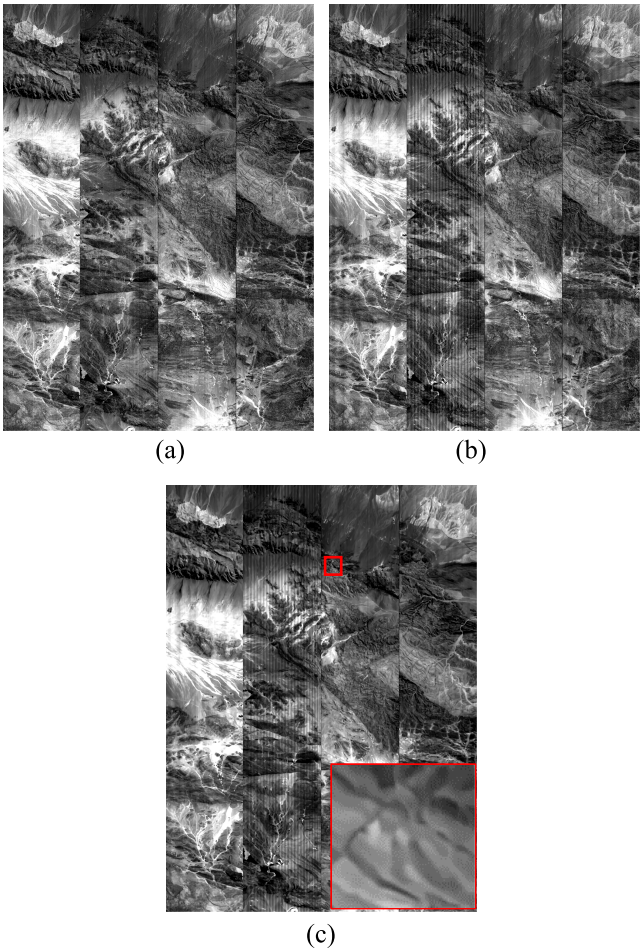

Fig. 15. East Tianshan data (band 160) acquired by AHSI. (a) Method proposed by Hu [10]. (b) LRMR method. (c) BM4D method.

[see the inset of Fig. 15(c)]. This is because these methods misjudge interference fringes as the texture information of the image.

To quantitatively evaluate the effectiveness of the proposed method, the image quality improvement factor I_{F2} [20] is used, and the results of different denoising methods are compared. I_{F2} is given by

$$I_{F2} = 10lg\left[\frac{\sum_x \left(m_{\text{DN}_f}[x] - m_{\text{DN}_f}[x-1]\right)^2}{\sum_x \left(m_{\text{DN}_r}[x] - m_{\text{DN}_r}[x-1]\right)^2}\right] \tag{17}$$

where m_{DN_f} is the mean value along the flight direction before correction and m_{DN_r} is the mean value along the flight direction after correction.

I_{F2} before and after correction (the data of East Tianshan) by three different proposed methods and the method used in this article in bands 94 and 160 are shown in Table II. Band 36 does not contain the interference effect, while bands 94 and 160 are deeply affected by the interference fringes. As it can be seen from Table II, comparing the new method with three previously proposed methods, I_{F2} of band 94 (160) has been improved from 0 to 2.83 (3.56), 5.72 (6.82), 19.77 (22.38), and 20.43 (22.43), respectively. It means that both the method

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TABLE II
IF2 BEFORE AND AFTER CORRECTION

Band	Raw	LRMR	BM4D	Paper[9]	Proposed
94	0	2.83	5.72	19.77	20.43
160	0	3.56	6.82	22.38	22.43

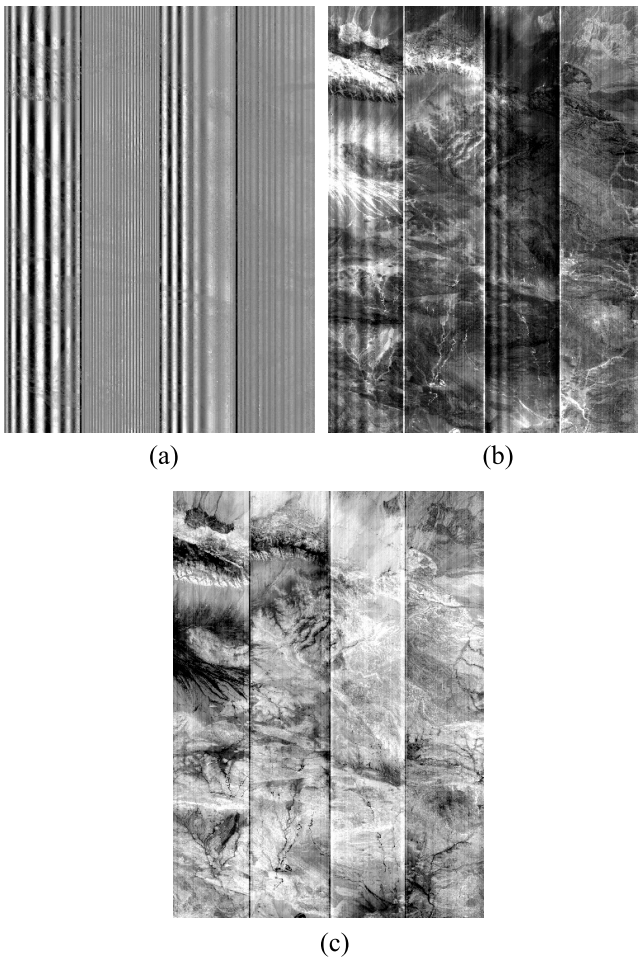

Fig. 16. Data from East Tianshan after MNF transformation (band 6). (a) Uncorrected hyperspectral image. (b) Corrected hyperspectral image using the method proposed by Hu *et al.* [10] (c) Corrected hyperspectral image using the method used in this article.

proposed by Hu *et al.* [10] and the method proposed in this article can greatly improve the quality of the image data.

Green [21] proposed the minimum noise fraction (MNF) transformation, which is used to determine the intrinsic dimension of image data and isolate noise in data. After MNF transformation, the transformed data space can be divided into two parts: one part is associated with large eigenvalues and the corresponding feature images, and the rest is associated with small eigenvalues and noise. And the data will be rearranged in the order from high signal-to-noise ratio (SNR) to low SNR. Interference fringes may reappear when the image is transformed into MNF space [22] or principal component analysis (PCA) space [23] even when they seem to be almost removed in the individual bands of the original image. Therefore, the effectiveness of the correction method can be better evaluated by the quality of each band after MNF transformation.

After MNF transformation (the data of East Tianshan), images in band 6 before correction and that after correction by the method proposed by Hu *et al.* [10] and the proposed method in this article are shown in Fig. 16(a)–(c), respectively. Other bands can be seen in Figs. S8 and S9 in Supporting Information. As it can be seen, the original data contain a large amount of streak noise, and the stripe patterns are different due to the different variation in the thickness of the Si layers among four IRFPAs of the detector. After the correction of the method proposed by Hu *et al.* [10], the image quality has improved, but a certain amount of fringes can still be observed. While after the correction of the method used in this article, the fringes of the four modules have been greatly suppressed, and only some nonuniformity of the pixels left and can be removed easily. Hu *et al.*'s method makes use of the spatial–spectral correlation of the adjacent bands of hyperspectral image. However, the spatial–spectral feature of data is not as strong as the normal hyperspectral data due to the influence of the etalon effect, which leads to some small interference fringes remaining after processing. These fringes are difficult to distinguish in the single-band image, but after MNF transformation, these residual noises will be highlighted. The proposed new method makes no assumptions on the spatial–spectral feature of hyperspectral data and makes full use of the statistical characteristics of the interference fringes in the spatial dimension. Therefore, the interference fringes in image have been removed more thoroughly.

V. CONCLUSION

A two-step method to remove fringes for hyperspectral SWIR imagers has been proposed and evaluated using the image data acquired by AHSI on China's GaoFen-5 Satellite. It provides satisfactory results in improving image quality and preserving the characteristics of the spectral curve. The results show that this method can reduce the peak value of spatial fringes from about ($\pm 15\%$) to about ($\pm 4\%$). In addition, the MNF results show that the interference fringes in the SWIR image has been removed more thoroughly than the previously proposed method by Hu *et al.* [10]. The proposed method is not specific to any instrument and makes no assumptions on any calibration data and can be used for other hyperspectral SWIR imagers.

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